

ADS 2001

**G1 - PriceCatcher (Studio : Malaysia
Campus)**

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Executive Summary

Purpose

Price differences across Malaysian districts affect household affordability and the usefulness of public price-monitoring systems. This report uses the PriceCatcher dataset to examine how supermarket chains, retail formats, geography, product categories, store density, and market concentration help explain local price disparities.

Methods

The analysis used approximately 20 million cleaned PriceCatcher price observations across 169 districts, 13 states and three federal territories, from April 2024 to February 2026. Prices were standardised using a price index, where 1.0 represents the national median benchmark for the same item category.

Price index is calculated as follows:

$$\text{Price Index} = \frac{\text{Observed Store Price}}{\text{National Median Price for the Same Item Category}}$$

Values below 1.0 indicate prices below the benchmark, while values above 1.0 indicate prices above it. The report compares price indices across major chains, product categories, retail formats, states, and districts. Store density was measured using the number of recorded stores in each district. Market concentration is measured using a premise-count HHI, where each recorded store is treated as a market participant and higher values indicate that local retail activity is concentrated among fewer stores. K-means clustering was also used to group product categories by promotion frequency and discount depth.

Findings

The clearest finding is that major supermarket chains have different overall price positions. Tesco, Eonsave, and Lotus generally sit below the national benchmark, while AEON, AEON Big, and Cold Storage sit above it. However, chain averages hide important product-level differences. A chain that is cheaper overall is not necessarily cheaper for every product, suggesting that retailers use product-specific pricing strategies rather than one uniform pricing approach.

Retail format also affects prices. Wholesale-style stores tend to sit below the benchmark, while smaller and more convenience-oriented formats such as Kedai Runcit and Pasar Basah tend to sit closer to or above it. This means consumers' price experiences depend not only on which chains operate nearby, but also on the types of stores available in their district.

Geography changes how prices are experienced. The comparison of prices in supermarket chain, AEON BIG, between Kuala Lumpur and Penang suggests that prices may vary by location, so national retailers should not automatically be assumed to apply one uniform pricing pattern across all regions.

Product categories also behave differently. Seasonal and perishable goods naturally show stronger price movement, while stable but high-priced categories may indicate fewer substitutes, stronger brand loyalty, or weaker competition. Price monitoring should therefore use category-specific expectations rather than a single threshold for all goods.

Store density and market concentration provide useful but limited evidence. Store density has a statistically significant negative relationship with price index, while HHI has a statistically significant positive relationship. However, both relationships have low R^2 values, meaning they explain only a small share of price variation. These indicators should be used as supporting context, not as standalone evidence of pricing risk.

Implications

Broad averages are not enough for effective price monitoring. Stakeholders should monitor prices at the chain, product-category, retail-format, and district levels. In practice, this means setting category-specific alert thresholds, comparing prices against local purchasing power, and using store density and market concentration as secondary indicators.

Limitations

The main limitation is that broad item categories can contain products with different brands, package sizes, and quality levels. Since the dataset contains too many SKUs to interpret individually in a client-facing report, the price index is a practical compromise for identifying broader patterns. The results should therefore be interpreted as evidence of broad retail price positioning rather than exact like-for-like SKU comparisons. Temporal recording is also inconsistent, with some items appearing intermittently across months.

1. Introduction

Price differences across Malaysian districts affect household affordability, consumer welfare, and the usefulness of public price-monitoring systems. National average prices can suggest stability while hiding meaningful differences between states, districts, store formats, retail chains, and product categories. This is especially important for essential goods such as chicken, eggs, cooking oil, vegetables, baby products, and fresh goods, where repeated purchases make even small price differences meaningful over time.

This report uses Malaysia's PriceCatcher dataset to examine how prices differ across retailers, products, store formats, and locations. OpenDOSM describes PriceCatcher transactional records as data collected for price surveillance and suitable for high-frequency analysis of specific items in specific locations, while noting that inflation analysis should still rely on CPI data rather than PriceCatcher alone (Department of Statistics Malaysia [DOSM], 2026). This report therefore does not estimate inflation, but instead uses PriceCatcher records to study local price variation.

The analysis focuses on major supermarket chains, including Tesco, Econsave, Lotus, AEON, AEON Big, and Cold Storage, because they operate across multiple locations and allow pricing patterns to be compared across products and regions. The report also examines retail format, geography, product-category behaviour, store density, and market concentration to understand the wider market conditions in which these prices occur.

The central argument is that Malaysian retail prices are shaped by overlapping market forces rather than one single driver. Chain-level pricing by supermarkets provides a useful starting point, but product-level pricing, retail format, geography, and local market structure all affect how consumers experience prices. The report begins with chain-level pricing patterns, then examines the wider market conditions that help explain price differences.

2. Data Overview, Data Quality, and Modelling Approach

2.1 Dataset overview

The PriceCatcher dataset contains transaction-level price observations for monitored goods across Malaysian states, districts, premises, store formats, and product categories. The original dataset contained approximately 22 million records. After removing incomplete rows and excluding incomplete months, the cleaned working dataset contained approximately 20 million price observations, covering 169 districts, 13 states and three federal territories, and the period from April 2024 to February 2026.

Each row represents an observed item price at a specific premise and date. As shown in Figure 1, the dataset includes product identifiers, store identifiers, location variables, product categories, store formats, and observed prices. These variables allow prices to be compared across products, retail chains, store formats, districts, and states.

Figure 2 shows that prices are strongly right-skewed. The median price is RM8.90, while the mean is approximately RM11.76, with values ranging from RM0.01 to RM6,399.00. This wide range reflects the fact that the dataset contains many different product types and package sizes. Raw prices are therefore not directly comparable across products, supporting the use of a standardised price index.

Figure 1

Preview of the cleaned PriceCatcher dataset.

	date	premise_code	item_code	price	Item	unit	address	premise_type	state	district	year	month	item_group	item_category	store_name
0	2/4/2025	2	1	10.0	AYAM BERSIH - STANDARD	1kg	JALAN LAKSAMANA,TAMAN JUBILEE,30300 IPOH, PERAK	Pasar Basah	Perak	Kinta	2025	4	Fresh Goods	Chicken	PASAR BESAR IPOH
1	2/4/2025	2	9	35.0	DAGING KAMBING BEBIRI IMPORT BERTULANG (MUTTON...	1kg	JALAN LAKSAMANA,TAMAN JUBILEE,30300 IPOH, PERAK	Pasar Basah	Perak	Kinta	2025	4	Fresh Goods	Meat	PASAR BESAR IPOH
2	2/4/2025	2	16	4.0	BETIK BIASA	1kg	JALAN LAKSAMANA,TAMAN JUBILEE,30300 IPOH, PERAK	Pasar Basah	Perak	Kinta	2025	4	Fresh Goods	Fruits	PASAR BESAR IPOH
3	2/4/2025	2	18	6.0	PISANG BERANGAN	1kg	JALAN LAKSAMANA,TAMAN JUBILEE,30300 IPOH, PERAK	Pasar Basah	Perak	Kinta	2025	4	Fresh Goods	Fruits	PASAR BESAR IPOH
4	2/4/2025	2	19	6.0	PISANG EMAS	1kg	JALAN LAKSAMANA,TAMAN JUBILEE,30300 IPOH, PERAK	Pasar Basah	Perak	Kinta	2025	4	Fresh Goods	Fruits	PASAR BESAR IPOH

Note. The table shows the first five rows of the working dataset, including product, store, location, date, and classification variables used in the analysis.

Figure 2

Descriptive statistics for numerical variables in the cleaned PriceCatcher dataset.

	date	premise_code	item_code	price	year	month
count	20147436	2.014744e+07	2.014744e+07	2.014744e+07	2.014744e+07	2.014744e+07
mean	2025-08-29 01:25:19.000235264	1.358072e+04	9.663570e+02	1.176394e+01	2.025147e+03	6.663763e+00
min	2025-04-02 00:00:00	2.000000e+00	1.000000e+00	1.000000e-02	2.025000e+03	1.000000e+00
25%	2025-06-06 00:00:00	5.274000e+03	1.130000e+02	4.990000e+00	2.025000e+03	4.000000e+00
50%	2025-08-18 00:00:00	1.709600e+04	1.131000e+03	8.900000e+00	2.025000e+03	7.000000e+00
75%	2025-11-21 00:00:00	2.081200e+04	1.561000e+03	1.420000e+01	2.025000e+03	9.000000e+00
max	2026-02-28 00:00:00	2.215300e+04	2.020000e+03	6.399000e+03	2.026000e+03	1.200000e+01
std	NaN	7.558195e+03	7.287210e+02	1.116684e+01	3.538576e-01	3.234190e+00

Note. The table reports count, mean, standard deviation, minimum, quartiles, and maximum values for key numerical variables, including `premise_code`, `item_code`, `price`, `year`, and `month`.

2.2 Data cleaning and preprocessing

Several preprocessing steps were applied before analysis to ensure that the working dataset was suitable for chain-level, product-level, geographic, and retail-format comparisons. Records with missing values in key analytical fields were removed, including product identifiers, observed price, premise information, store type, state, district, date, and store name. Complete-case deletion was considered appropriate because the remaining dataset was still large, with approximately 20 million observations retained after cleaning, and continued to cover 169 districts across 13 states and three federal territories. This means approximately 10% of records were removed during cleaning.

Missing values were not imputed because most of the affected fields represented product, store, or location identities. Imputing these variables would require assigning artificial products, retailers, districts, or premise types, which could distort the comparisons made later in the report.

Price values were also reviewed for extreme observations. Raw prices were strongly right-skewed because the dataset contains many different products, units, package sizes, and quality levels. Extreme raw prices were therefore not automatically removed. Instead, the analysis relied mainly on medians and standardised price indices, which reduce the influence of unusually high or low raw prices. This approach is more appropriate than comparing raw mean prices across products such as chicken, eggs, cooking oil, vegetables, and baby products, which naturally have different price levels.

Repeated observations were also considered. In a high-frequency price-monitoring dataset, repeated item-store-date observations can occur because the same item may be recorded across many stores and dates. These records were treated as valid monitoring observations rather than errors. For this reason, the analysis places more emphasis on medians, aggregated price indices, and district-level summaries than on raw record counts alone.

2.3 Price index construction and feature engineering

Raw prices are not directly comparable across products because different goods have different natural price levels. For example, a kilogram of chicken, a tray of eggs, and a tin of baby formula cannot be meaningfully compared using raw prices alone. To address this, prices were standardised using a price index, where the price index is defined as follows:

$$\text{Price Index} = \frac{\text{Observed Store Price}}{\text{National Median Price for the Same Item Category}}$$

A price index of 1.0 represents the national market benchmark, defined here as the national median price for the same item category. Values below 1.0 indicate prices below the benchmark, while values above 1.0 indicate prices above the benchmark. This index allows prices to be compared across chains, product categories, retail formats, districts, and states.

The price index is also a methodological compromise. Ideally, prices would be compared at the individual SKU level, because products within the same category can differ by brand, package size, quality tier, and product type. However, the dataset contains a very large number of monitored items, making SKU-by-SKU interpretation impractical for a client-facing report. The analysis therefore uses category-level and aggregated price indices to identify broader pricing patterns. This improves interpretability, but it also means that some observed differences may reflect product-mix differences rather than pure like-for-like price differences.

Feature engineering was used to create the variables required for the main comparisons. Store counts were calculated at the district level to measure local store density. Retail-format shares were calculated to compare the mix of premise types across states and districts. Prices were also aggregated by chain, item category, retail format, district, state, and month depending on the research question. These aggregations allowed the analysis to examine price variation at several levels: overall supermarket chain positioning, product-specific pricing niches, retail-channel differences, geographic differences, category-level pricing behaviour, and district-level market structure.

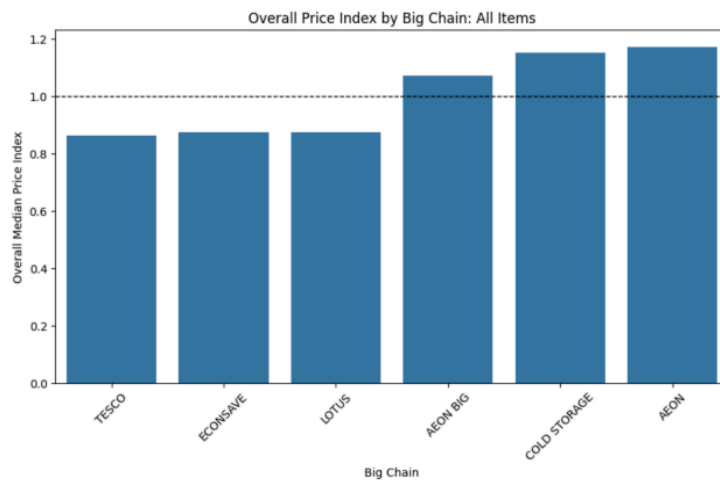
The modelling approach was designed to be interpretable and client-facing. The report first compares price indices across chains, product categories, retail formats, and geography. K-means clustering is then used to group product categories with similar promotion frequency and discount depth. Finally, correlation and fitted trend-line analysis are used to examine whether store density and premise-count HHI are associated with district-level price differences. This approach is appropriate because the main objective is to identify pricing patterns and monitoring priorities, rather than to predict every individual item price.

3. Do Major Supermarket Chains Have Different Overall Price Positions?

The first stage of the analysis compares major supermarket chains using the price index. This gives an overall view of whether each chain tends to price below or above the national market benchmark, defined here as the national median price for the same item category. The chains examined are Tesco, Econsave, Lotus, AEON, AEON Big, and Cold Storage.

Figure 3

Overall price index by major supermarket chains.



Note. The bar chart compares each major chain's average price index against the national market benchmark of 1.0.

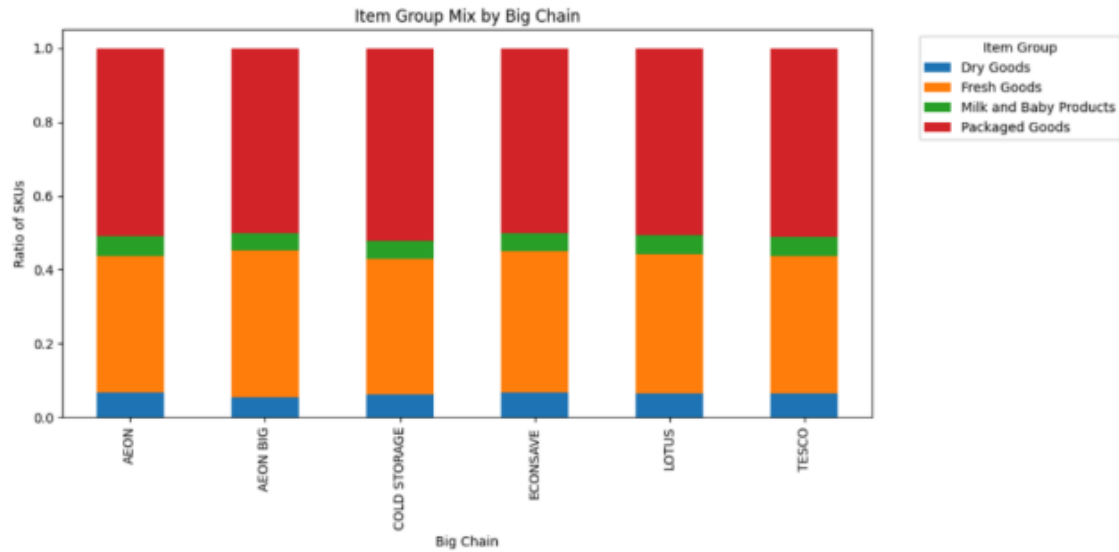
Figure 3 shows that Tesco, Econsave, and Lotus sit below the national benchmark, while AEON, AEON Big, and Cold Storage sit above it. This indicates that the major chains do not share the same overall price position. Some chains appear more discount- or affordability-positioned, while others appear closer to a premium price position.

This interpretation is consistent with retail research showing that supermarkets and food retailers can follow different pricing strategies even within the same market. Anania and Nisticò (2014) found that price dispersion in food retail can remain high even for homogeneous products and that retailers may differ in pricing strategies, including supermarkets belonging to the same chain. Retail positioning research also shows that retailers often compete through different price positions and category strategies rather than through a single uniform price approach (González-Benito & Martos-Partal, 2012).

However, the overall price index only reflects the average price position of the chain across the full basket of observed products. It does not show which product categories each chain is cheaper or more expensive in. Therefore, the next step is to check whether the chain-level differences are simply caused by different product mixes.

Figure 4

Item group mix by major supermarket chains.



Note. The stacked bar chart compares the proportion of item groups sold across major supermarket chains.

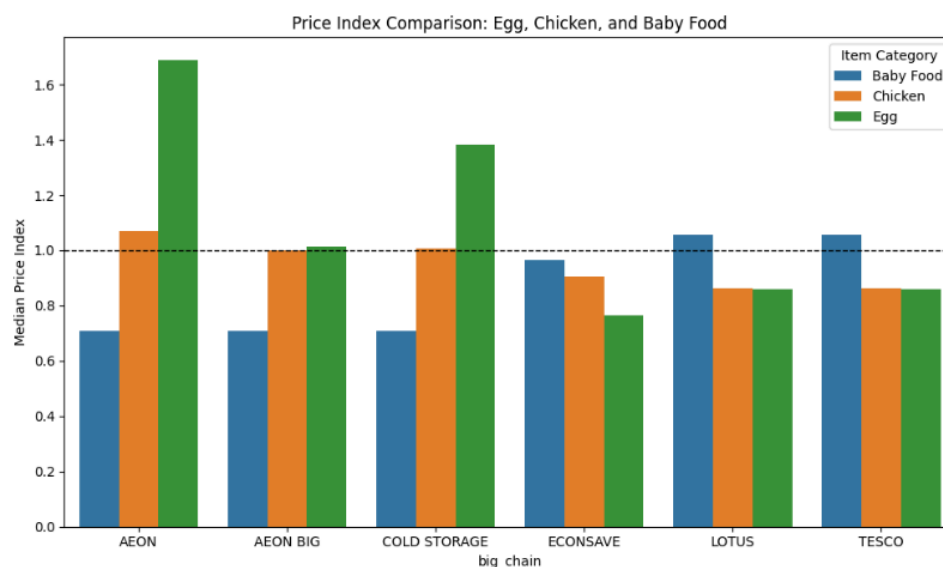
Figure 4 shows that item-group composition was broadly similar across major chains. Packaged goods formed the largest share for every chain, while dry goods, fresh goods, and milk and baby products appeared in relatively consistent proportions. This means that overall chain price differences are unlikely to be explained only by one chain selling a completely different mix of products. Instead, the evidence supports the interpretation that chain-level market positioning partly explains pricing differences.

4. Do Supermarket Chain Averages Hide Product-Specific Pricing Niches ?

The overall supermarket chain price index gives a useful starting point, but it cannot show how each chain prices individual product categories. A chain may be cheaper overall while still being expensive for certain products, or more expensive overall while still competing aggressively on selected categories. To examine this, Figures 5 and 6 compare price indices at the product-category level.

Figure 5

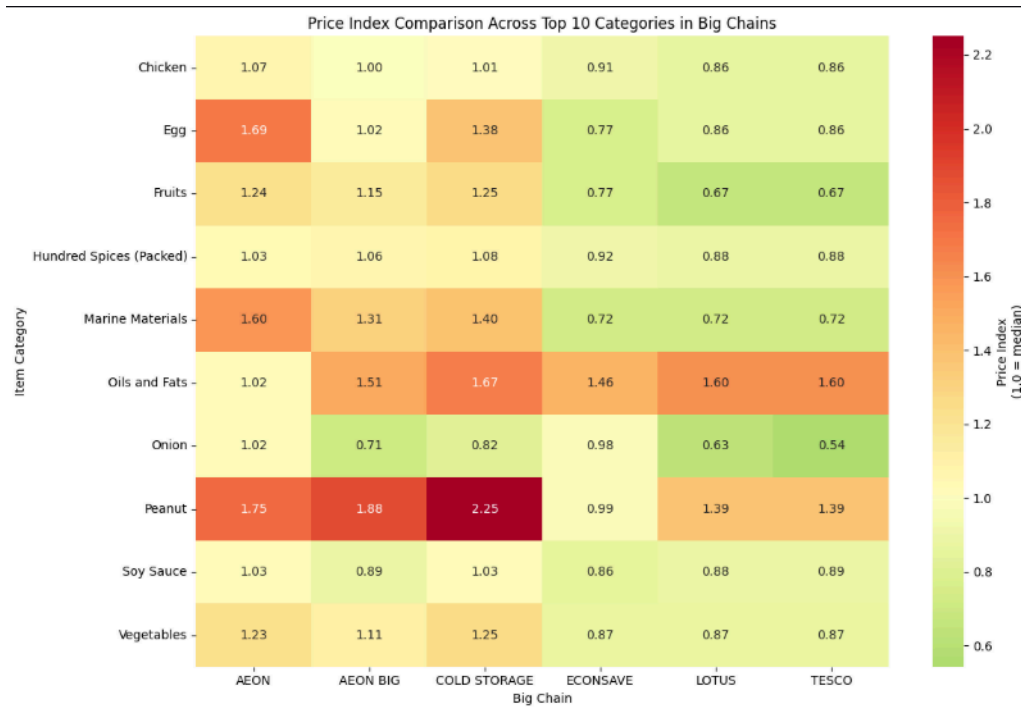
Product-specific price index by major retail chains.



Note. The grouped bar chart compares chain-level price indices across selected product categories.

Figure 6

Price index comparison across top 10 categories in major chains.



Note. The heatmap compares price indices across major supermarket chains and selected item categories. Values above 1.0 indicate prices above the national benchmark, while values below 1.0 indicate prices below the benchmark.

Figures 5 and 6 show that chain pricing varies by product category. A chain that appears discount-positioned overall is not necessarily below the benchmark for every product. Similarly, a chain that appears premium-positioned overall may still offer competitive prices for selected categories. This confirms that chain averages hide product-specific pricing niches.

Figure 5 illustrates this point using selected essential categories such as baby food, chicken, and eggs. For example, some discount-positioned chains remain below the benchmark for chicken and eggs, while other chains are above the benchmark for the same items. Figure 6 extends this pattern across a wider range of categories, including chicken, eggs, fruits, marine materials, oils and fats, onions, peanuts, soy sauce, and vegetables. The heatmap makes the category-level differences clearer because it shows where each chain is above or below the benchmark across many products at once.

This supports the idea that supermarket chains compete through category strengths and weaknesses, rather than through one uniform pricing strategy across the entire store. For instance, a chain may price chicken or eggs attractively to draw regular household shoppers, while maintaining higher prices in categories such as oils and fats or peanuts. Another chain may appear premium overall but still compete strongly on selected staple products.

This interpretation is consistent with multiproduct retailing theory. Supermarkets do not sell single goods in isolation; they sell baskets of goods. Walsh and Whelan (1999) explain that multiproduct retailers may use selected items as “traffic builders,” pricing them aggressively to attract consumers while recovering margins elsewhere. Hosken and Reiffen (2007) similarly argue that discounts on individual products can influence consumers’ store choice because consumers often buy bundles of goods from the same retailer. This is consistent with a possible loss-leader interpretation, where selected items are priced competitively to attract customers while profits are recovered across the wider basket.

Retail positioning research also shows that both retailer position and product category matter when analysing retail behaviour, meaning store-wide averages can miss important category-level differences (González-Benito & Martos-Partal, 2012). Evidence from supermarket scanner data also suggests that margins can fall on selected products during high-demand periods, which is consistent with retailers using particular goods strategically rather than pricing every category uniformly (Chevalier et al., 2003).

Therefore, the key insight is not that one chain is simply cheap and another is simply expensive. A more accurate conclusion is that each chain appears to have product categories where it is more competitive and others where it is less competitive.

5. Do Different Retail Channels Have Different Price Levels?

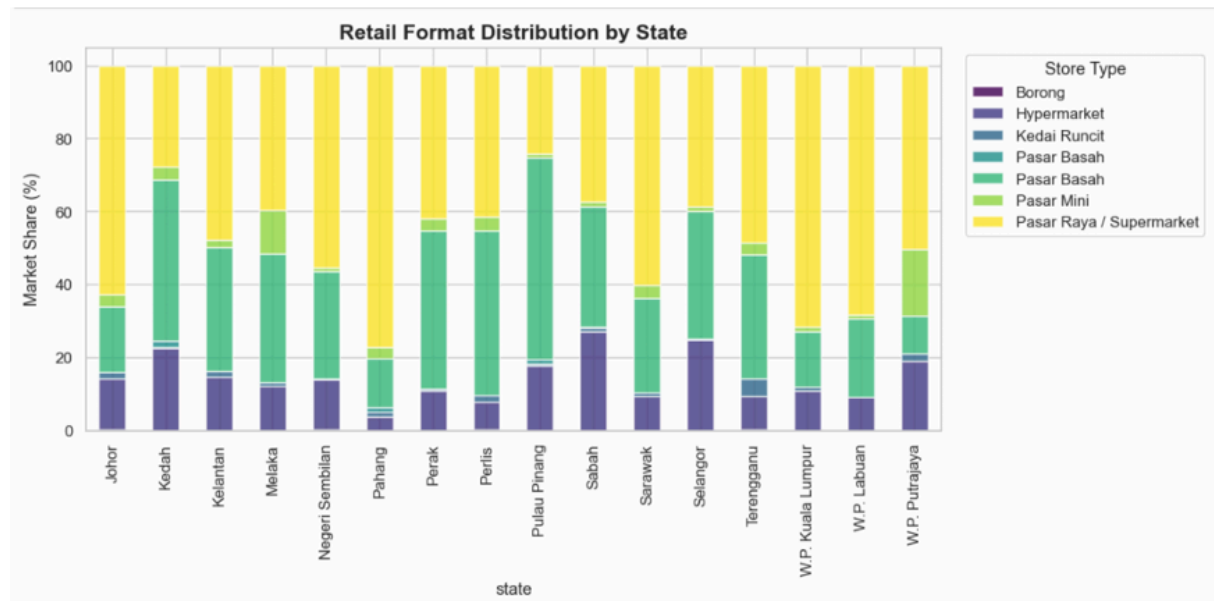
The supermarket chain-level and product-level analyses suggest that retailer strategy contributes to price differences across districts, states, and regions. However, chain strategy is only part of the retail environment. A consumer’s price experience also depends on the type of store available to them. In the PriceCatcher dataset, the main retail channels are Borong (wholesale stores), Hypermarkets, Kedai Runcit (small grocery stores), Pasar Basah (wet markets), Pasar Mini (mini markets), and Pasar Raya / Supermarkets.

Before comparing price levels by retail channel, it is useful to examine how retail formats are distributed across states. If different store types have different price positions, then differences in retail-format composition may help explain why some states or districts experience different price levels.

In this section, retail-channel composition refers to the mix of store types recorded in each state or district. This is one part of market structure because consumers may face different prices depending on whether they mainly shop at supermarkets, wet markets, small grocery stores, hypermarkets, or wholesale stores. If each retail format has a different price level, then the format mix in a state or district may help explain the prices consumers observe.

Figure 7

Retail format distribution by state.



Note. The stacked bar chart compares the market share of each retail format across Malaysian states and federal territories. Retail formats include Borong, Hypermarket, Kedai Runcit, Pasar Basah, Pasar Mini, and Pasar Raya / Supermarket.

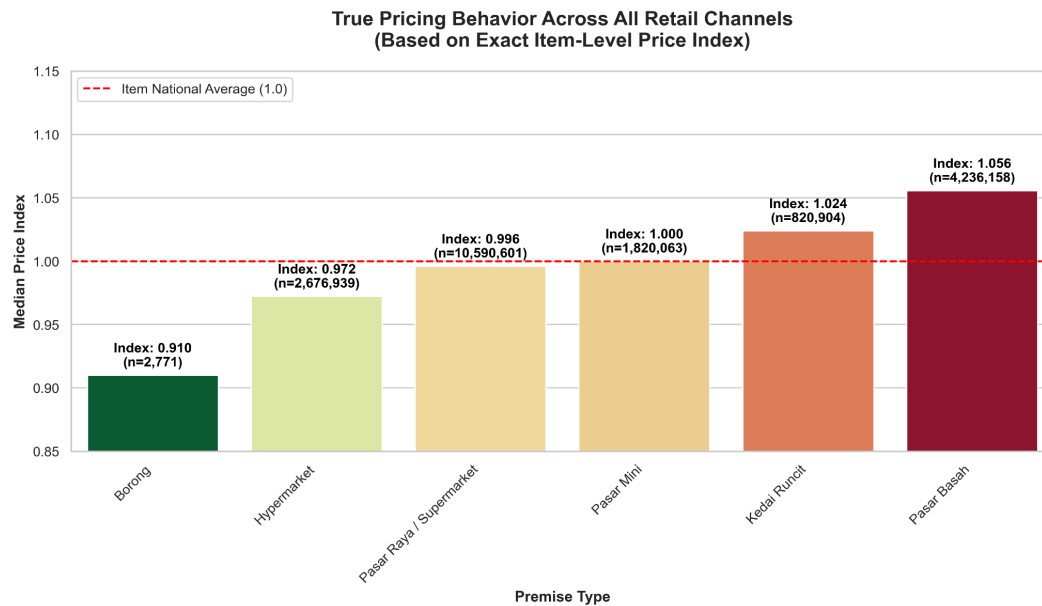
Figure 7 shows that Pasar Raya / Supermarket is the dominant recorded retail format in most states in the PriceCatcher dataset. This suggests supermarkets are a central retail channel for observed price monitoring. However, the extent of supermarket dominance differs by state. Some states show a stronger presence of wet markets or small grocery stores, while others have a larger share of supermarkets and hypermarkets.

This matters because retail channels are not interchangeable. A supermarket, wet market, small grocery store, and wholesale store may sell overlapping goods, but they operate with different business models. Wholesale stores may compete more on bulk or low-cost pricing. Hypermarkets and supermarkets may benefit from scale, standardised sourcing, and wider product range. Small grocery stores may charge higher prices because they provide convenience and neighbourhood access. Wet markets may have different prices because they often sell fresh and perishable goods, where prices can change depending on daily supply and product freshness. These differences mean that state or district price levels may partly depend on the retail channels available to consumers.

Figure 8 compares price levels across these retail formats using the price index.

Figure 8

Median price index across retail premise types.



Note. The bar chart compares median price index values across retail formats against the national benchmark of 1.0.

Figure 8 shows a clear price ladder across retail formats. Borong (wholesale-style stores) sit below the national benchmark, suggesting that they are generally more affordability-oriented in the observed data. Hypermarkets and Pasar Raya / Supermarkets are closer to the benchmark. Smaller or more convenience-oriented formats, such as Kedai Runcit (small grocery stores), and market-based formats such as Pasar Basah (wet markets), tend to sit closer to or above the benchmark.

This supports the idea that retail format is a structural contributor to price differences. Price differences are not only about which chain is operating in a state or district; they are also shaped by the kind of retail channel consumers rely on. A state where supermarkets and wholesale outlets are more accessible may have a different price environment from a state where consumers rely more heavily on small grocery stores or wet markets. This is consistent with food retail research showing that price dispersion can persist because sellers differ in format, location, assortment of goods sold, and customer base (Anania & Nisticò, 2014).

The key insight is that retail-channel composition matters. Since different formats have different price levels, the mix of store types in a state or district may influence the prices consumers observe. This leads naturally into the next question: if retail formats differ by state, do chain prices and purchasing power also differ geographically?

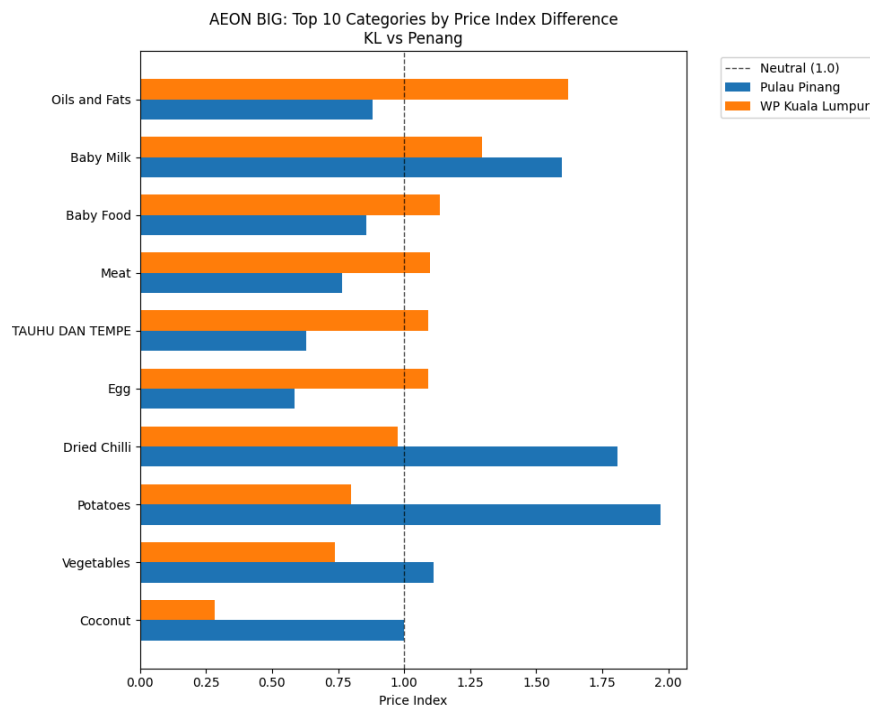
6. Does Geography Change How Chain Prices Are Experienced?

The previous section showed that retail formats differ in both distribution and price level. This raises a related geographic question: if states have different retail-channel mixes, do consumers also experience different chain prices and affordability pressures across regions?

This section examines geography by using supermarket chain, AEON Big, as an example to compare selected goods across Kuala Lumpur and Penang. This comparison shows whether the same chain displays similar or different price-index patterns across locations.

Figure 9

Regional price index comparison between Kuala Lumpur and Penang.



Note. The grouped bar charts compare Aeon Big price indices across Kuala Lumpur and Penang.

Figure 9 shows that AEON Big has different price-index patterns across Kuala Lumpur and Penang. This suggests that at least some chain-level prices may vary by location. For example, selected products at AEON Big may appear cheaper in Penang than in Kuala Lumpur, while other products may show a smaller difference.

This finding is useful because it challenges the idea that a chain has one fixed national price identity. A chain may be generally affordability-positioned overall, but its pricing can still vary depending on local costs, store location, regional competition, supply conditions, and demand patterns. This is consistent with

the idea that prices for similar goods may not fully converge across locations when markets are affected by transport costs, local competition, and other regional frictions (Allington et al., 2005).

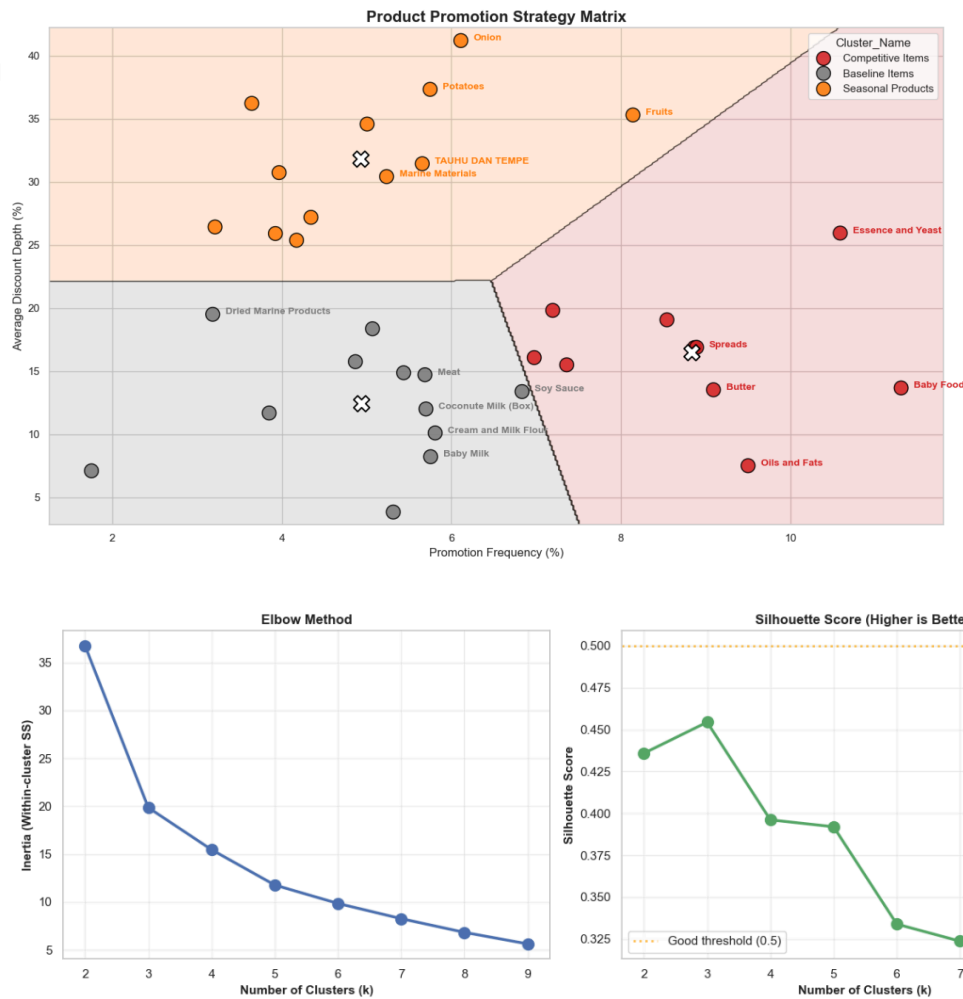
7. Which Product Categories Show Stronger Price Variation?

The previous sections showed that prices differ by chain, product category, retail format, and geography. A related question is whether some product categories naturally show stronger price variation than others. This matters because price movement is not always a warning sign. Seasonal or perishable products may fluctuate because of supply conditions, while stable but high-priced products may raise different concerns, such as fewer substitutes or weaker competition.

This section uses a promotion frequency and discount depth matrix to classify product-level price behaviour. Promotion frequency measures how often a product category experiences price reductions, while discount depth measures how large those reductions are when they occur. K-means clustering was used as a segmentation method, with $k = 3$ chosen based on the silhouette score, elbow plot, and interpretability.

Figure 10

K-means promotion strategy matrix and cluster diagnostics



Note. The main scatter plot classifies product categories by promotion frequency and average discount depth using K-means clustering with $k = 3$. Categories are grouped into baseline items, seasonal products, and competitive items. The elbow and silhouette plots are included to find the optimal number of clusters (k).

The number of clusters was selected using both interpretability and clustering diagnostics. As shown in Figure 10, the silhouette score is highest when $k = 3$, indicating that the three-cluster solution provides the clearest separation among the tested options. The elbow plot also shows that within-cluster inertia falls sharply at lower values of k before improvements become more gradual. Therefore, $k = 3$ was retained because it provides a simple, interpretable, and diagnostically supported segmentation of product-category pricing behaviour.

Figure 10 separates product categories into three broad groups: baseline items, seasonal products, and competitive items. This distinction is important because the same percentage price movement does not

mean the same thing for every product category. A large discount or price movement may be normal for seasonal and perishable goods, but the same movement in a normally stable category may be a stronger warning signal. This is consistent with evidence that regular prices and temporary sale prices behave differently, meaning short-term price movements should be interpreted relative to the normal pricing behaviour of each product category (Nakamura & Steinsson, 2008).

The baseline items cluster contains categories with relatively low promotion frequency and lower discount depth. These products appear to have more stable pricing behaviour in the observed data, meaning large discounts or sudden price movements are less common. For these categories, a major change in either promotion frequency or discount depth should attract attention because it represents a larger departure from normal behaviour. For example, if a normally stable category such as baby-related products suddenly shows a deep discount or a sustained price increase, this may indicate a meaningful change in retailer strategy, supply conditions, or affordability pressure.

The seasonal products cluster contains categories with deeper discounts but less frequent promotions. These products are more likely to be affected by seasonality, perishability, supply conditions, and short-term stock movements. Categories such as fruits, onions, potatoes, and marine-related products can experience larger price changes when supply conditions shift. For these products, a large movement in discount depth is not automatically concerning because volatility is part of their normal pricing behaviour. For example, a 30% change in a seasonal or perishable category may be less surprising than the same change in a stable household staple. This is consistent with research showing that promotional demand and discount behaviour for perishable products can vary depending on product characteristics and shelf-life conditions (van Donselaar et al., 2016).

The competitive items cluster contains categories with higher promotion frequency but generally less extreme discount depth than seasonal products. These products appear to be promoted more regularly, suggesting that retailers may use them to compete for household shoppers. However, because the discounts are more frequent and less unusual, they should not automatically be interpreted as abnormal price instability. This interpretation is consistent with multiproduct retailing theory, where retailers may discount selected items to influence store choice because consumers often buy baskets of goods rather than isolated products (Hosken & Reiffen, 2007; Walsh & Whelan, 1999). Therefore, this cluster is best understood as evidence of regular competitive promotional behaviour.

The practical value of Figure 10 is that it provides a category-specific baseline for price monitoring. Stakeholders should not apply one fixed alert threshold across all product categories. For seasonal products, larger price or discount movements may be normal and should only raise concern when they are unusually persistent or much larger than historical behaviour. For baseline items, even smaller changes may deserve closer attention because these products are normally stable. For competitive items, frequent promotions may reflect normal retailer competition rather than unusual pricing behaviour. In short, the matrix helps distinguish normal category behaviour from price movements that may require closer monitoring.

8. Do Store Density and Market Concentration Explain Price Differences?

After examining chain strategy, product-level price behaviour, retail format, and geography, the next question is whether price differences are associated with simple competition indicators. In theory, stronger competition should reduce prices because consumers have more alternatives and retailers have less pricing power. However, food retail markets can still show price dispersion because retailers differ in format, location, assortment, and customer base (Anania & Nisticò, 2014). Spatial retail competition also depends on whether stores are realistic substitutes for consumers, not only on how many stores are recorded in an area (Ellickson et al., 2020).

This section tests two district-level competition indicators: store density and market concentration. Store density refers to the number of stores recorded in a district. It is used as a simple indicator of how many retail options consumers appear to have. Market concentration is measured using an adapted premise-count Herfindahl-Hirschman Index (HHI). The standard HHI is calculated by squaring each market participant's share and summing the values, with higher values indicating greater concentration (U.S. Department of Justice, 2024).

In this report, each recorded store is treated as a separate market participant for the purpose of calculating HHI. This means the measure captures how concentrated or fragmented the recorded store base is within each district. A higher HHI indicates that recorded market activity is more concentrated among fewer stores, while a lower HHI indicates a more fragmented local retail environment. However, this is still a proxy for market power because the dataset does not contain firm-ownership information. The objective is to test whether districts with fewer or more concentrated retail options tend to have higher price-index values.

Figure 11

Average price index versus number of stores.



Note. The scatter plot compares district-level average price index against the number of stores recorded in each district. The fitted trend line shows the overall relationship between store density and price index.

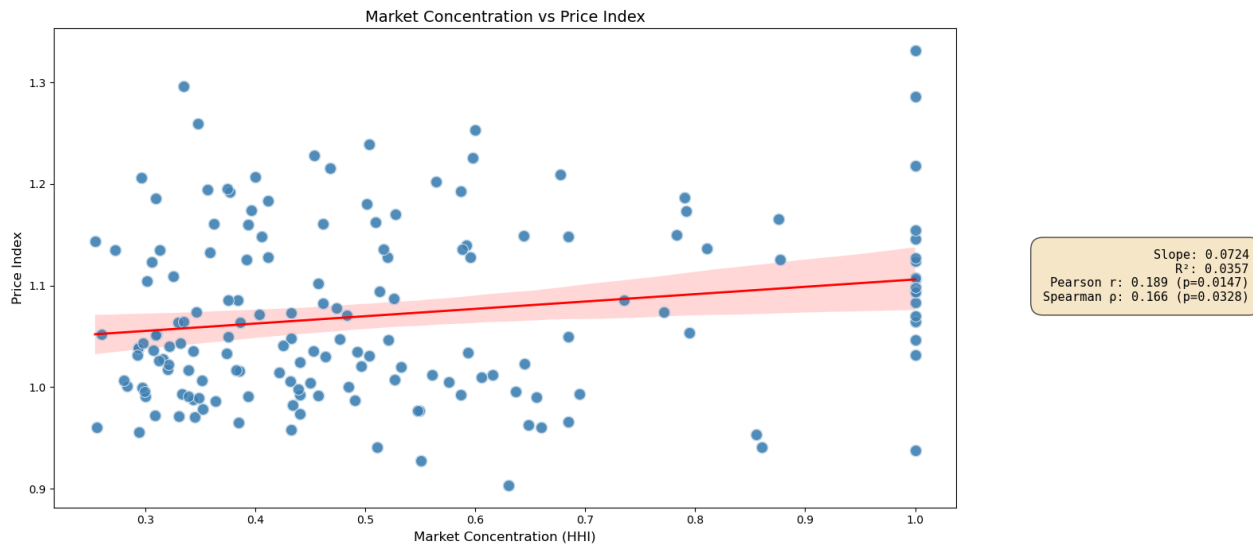
Figure 11 examines the relationship between district-level average price index and the number of recorded stores. The scatter plot shows a negative association: districts with more recorded stores tend to have lower average price-index values. This is supported by the fitted regression line and the correlation results. The Pearson correlation is -0.333 and the Spearman correlation is -0.393, with both p-values close to zero, indicating that the relationship is statistically significant in the dataset.

The R^2 value is 0.111, meaning that store count accounts for about 11.1% of the observed variation in district-level price index. While this is not a large share, it is still useful in context. Store density appears to capture one part of local competition: districts with more recorded stores generally show some downward pressure on prices. This supports the expectation that greater retail availability can improve consumer choice and reduce pricing pressure.

However, store density is most useful as a screening indicator rather than a complete explanation. The scatter shows that districts with similar store counts can still have different price-index values. This is expected because food retail competition depends not only on the number of stores, but also on store format, location, assortment, and customer base (Anania & Nisticò, 2014). Spatial retail competition also depends on whether competitor stores are realistic substitutes for consumers, not only on the total number of stores recorded in an area (Ellickson et al., 2020). Therefore, districts with low store density and high price-index values may be useful priority areas for closer monitoring.

Figure 12

Market concentration and district price index.



Note. The scatter plot compares district-level premise-count Herfindahl-Hirschman Index against district-level average price index. Each point represents one district. The fitted line shows the overall relationship between market concentration and the price index.

Figure 12 examines the relationship between market concentration and district-level price index. The fitted line shows a weak positive relationship, suggesting that districts with more concentrated recorded retail activity tend to have slightly higher price indices. The Pearson correlation is 0.189 with a p-value of 0.0147, while the Spearman correlation is 0.166 with a p-value of 0.0328. This indicates that the relationship is statistically significant, although smaller than the store-density relationship.

The R^2 value is 0.0357, meaning that market concentration accounts for about 3.6% of the observed variation in district-level price index. This shows that concentration is not the main driver of district price differences, but it still provides useful market-structure information. Districts with higher HHI values may have fewer effective retail alternatives, which can make them more vulnerable to higher prices, especially when combined with low store density, limited retail-format diversity, or geographic isolation.

In this report, each recorded store is treated as a separate market participant for the purpose of calculating HHI. The standard Herfindahl-Hirschman Index is calculated by squaring each market participant's share and summing the values, with higher values indicating greater concentration (U.S. Department of Justice, 2024). Since the dataset does not contain firm-ownership information, the HHI used here is best interpreted as a proxy for local market power.

For price-monitoring purposes, the most important districts are those where store density is low, market concentration is high, and the price index is above the benchmark. These districts are more likely to require closer investigation because consumers may face both higher prices and fewer local retail alternatives.

9. Conclusion and Client Implications

This report investigated how major retail chains and local market conditions explain price disparities across Malaysian districts. The analysis examined chain-level pricing, product-specific pricing niches, retail formats, geography, product-category variation, store density, and market concentration.

The clearest finding is that major supermarket chains have different overall price positions. Tesco, Econsave, and Lotus generally sit below the national benchmark, while AEON, AEON Big, and Cold Storage sit above it. However, product-level analysis shows that these averages hide important category differences. A chain that appears cheaper overall is not necessarily cheaper for every product, and a premium-positioned chain may still compete strongly in selected categories. This suggests that retailers use product-specific pricing strategies rather than applying one uniform pricing approach across the whole store.

Retail format and geography also affect how consumers experience prices. Wholesale-style stores tend to sit below the benchmark, while smaller or convenience-oriented formats such as Kedai Runcit and Pasar Basah tend to sit closer to or above it. The AEON Big comparison also shows that the same chain can display different price-index patterns across regions, suggesting that national retailers should not automatically be assumed to price uniformly across locations.

Product-level variation provides useful monitoring context. Seasonal and perishable goods may naturally show stronger price movement, while stable but high-priced categories may indicate fewer substitutes, stronger brand loyalty, or weaker competition. Price changes should therefore be assessed against the normal behaviour of each product category rather than using one threshold for all goods.

Store density and market concentration are useful but limited indicators. Store density has a statistically significant negative relationship with price index, while premise-count HHI proxy has a statistically significant positive relationship. However, both relationships have low R^2 values, meaning they explain only a small share of price variation. These indicators should therefore be used as supporting context, not as standalone evidence of pricing risk.

The main client implication is that broad averages are not enough for effective price monitoring. Stakeholders should monitor prices at the chain, product-category, retail-format, and local-market levels. In practice, this means setting category-specific alert thresholds, and using store density and concentration as secondary indicators alongside chain presence, product mix, and geography.

Overall, Malaysian retail prices are shaped by overlapping forces rather than one single driver. A useful monitoring system should therefore identify where retailer positioning, product-specific behaviour, retail format, and local market structure combine to create affordability pressure for consumers.

References

- Allington, N. F. B., Kattuman, P. A., & Waldmann, F. A. (2005). One market, one money, one price? *International Journal of Central Banking*, 1(3), 73–115.
<https://www.jbs.cam.ac.uk/wp-content/uploads/2020/08/wp0501.pdf>
- Anania, G., & Nisticò, R. (2014). Price dispersion and seller heterogeneity in retail food markets. *Food Policy*, 44, 190–201. <https://doi.org/10.1016/j.foodpol.2013.12.004>
- Chevalier, J. A., Kashyap, A. K., & Rossi, P. E. (2003). Why don't prices rise during periods of peak demand? Evidence from scanner data. *American Economic Review*, 93(1), 15–37.
<https://doi.org/10.1257/000282803321455142>
- Department of Statistics Malaysia. (2026). *PriceCatcher: Transactional records*. OpenDOSM.
<https://open.dosm.gov.my/data-catalogue/pricecatcher>
- Ellickson, P. B., Grieco, P. L. E., & Khvastunov, O. (2020). Measuring competition in spatial retail. *RAND Journal of Economics*, 51(1), 189–232. <https://doi.org/10.1111/1756-2171.12310>
- González-Benito, Ó., & Martos-Partal, M. (2012). Role of retailer positioning and product category on the relationship between store brand consumption and store loyalty. *Journal of Retailing*, 88(2), 236–249.
<https://doi.org/10.1016/j.jretai.2011.05.003>
- Hosken, D., & Reiffen, D. (2007). Pricing behavior of multiproduct retailers. *The B.E. Journal of Theoretical Economics*, 7(1), Article 39. <https://doi.org/10.2202/1935-1704.1354>
- Nakamura, E., & Steinsson, J. (2008). Five facts about prices: A reevaluation of menu cost models. *Quarterly Journal of Economics*, 123(4), 1415–1464. <https://doi.org/10.1162/qjec.2008.123.4.1415>
- U.S. Department of Justice. (2024). *Herfindahl-Hirschman Index*. Antitrust Division.
<https://www.justice.gov/atr/herfindahl-hirschman-index>
- van Donselaar, K. H., Peters, J., de Jong, A., & Broekmeulen, R. A. C. M. (2016). Analysis and forecasting of demand during promotions for perishable items. *International Journal of Production Economics*, 172, 65–75. <https://doi.org/10.1016/j.ijpe.2015.10.022>
- Walsh, P. P., & Whelan, C. (1999). Loss leading and price intervention in multiproduct retailing: Welfare outcomes in a second-best world. *International Review of Law and Economics*, 19(3), 333–347.
[https://doi.org/10.1016/S0144-8188\(99\)00015-0](https://doi.org/10.1016/S0144-8188(99)00015-0)

Appendix

Appendix A: Data dictionary and preprocessing details

Variable	Data type	Description	Role in analysis
date	datetime64[ns]	Date of the recorded price observation. Originally loaded as an object/string and converted using <code>pd.to_datetime()</code> .	Used for time-based grouping, monthly trends, volatility analysis, and promotion-related calculations.
premise_code	int64	Unique numerical identifier for each recorded store or premise.	Used to identify premises and calculate store counts by district or store type.
item_code	int64	Unique numerical identifier for each monitored product item.	Used to group products and support price comparison across stores and locations.
price	float64	Observed selling price of the item in Malaysian Ringgit.	Main price variable used to calculate the price index, average prices, volatility, and discount depth.
item	object	Name of the monitored product item.	Used to interpret product-level and category-level pricing patterns.

unit	object	Measurement unit for the item, such as 1kg, packet, bottle, or unit.	Provides context for interpreting the recorded price.
address	object	Address of the recorded premise.	Used as supporting location information.
premise_type	object	Retail format of the premise, such as hypermarket, supermarket, wet market, mini market, grocery store, or wholesale store.	Used to compare price levels across retail formats and analyse store type composition.
state	object	Malaysian state or federal territory where the premise is located.	Used for state-level geographic comparison and urban-rural classification.
district	object	District where the premise is located.	Used for district-level market structure analysis, store density, and market concentration.
year	int64	Year extracted from the date variable.	Used for filtering and temporal aggregation.

month	int64	Month extracted from the date variable.	Used for monthly grouping, volatility analysis, and promotion-related calculations.
item_group	object	Broad product group, such as fresh goods, dry goods, packaged goods, or milk and baby products.	Used to compare product mix across major supermarket chains.
item_category	object	More specific product categories, such as chicken, eggs, fruits, vegetables, oils and fats, or baby food.	Used to calculate category-level price indices and analyse product-specific pricing behaviour.
store_name	object	Store or premise name.	Used to identify and compare major supermarket chains such as Tesco, Econsave, Lotus, AEON, AEON Big, and Cold Storage.

Appendix B : Group Project Assessment for ADS2001

Group Project Assessment for ADS2001
Studio: Malaysia
Group: 1 PriceCatcher

		Lim Kee Yang	Jonathan Yek	Hanji e Yu	Ziyue Meng	Wenqi Yang	Koay Ji Wei	
Coordination	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	20	16	16	16	16	16	
Research	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	16	16	16	20	16	16	
Coding	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	16	16	16	16	20	16	
Report Writing	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	16	16	20	16	16	16	
Presentation Preparation	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	16	20	16	16	16	16	
Attendance at Final Presentation	This entry should be 1 (present) or 0 (absent)	1	1	1	1	1	1	

Appendix C : Acknowledgement of AI use

We acknowledge the use of ChatGPT (<https://chat.openai.com/>) in the preparation of this assignment. We used ChatGPT to refine the academic tone and accuracy of our writing, including grammatical structure, punctuation, and vocabulary, as well as to suggest improvements to the organisation and structure of the essay. All outputs were critically reviewed, edited, and adapted, and the final submission reflects our understanding, analysis, and writing style.